

GSx100 than the metrics themselves. Despite the correlation coefficients being relatively low, this is nonetheless a promising result. Evaluating goaltending is notoriously difficult, and a stable and predictive metric that does not rely on the actual number of goals allowed may prove valuable.

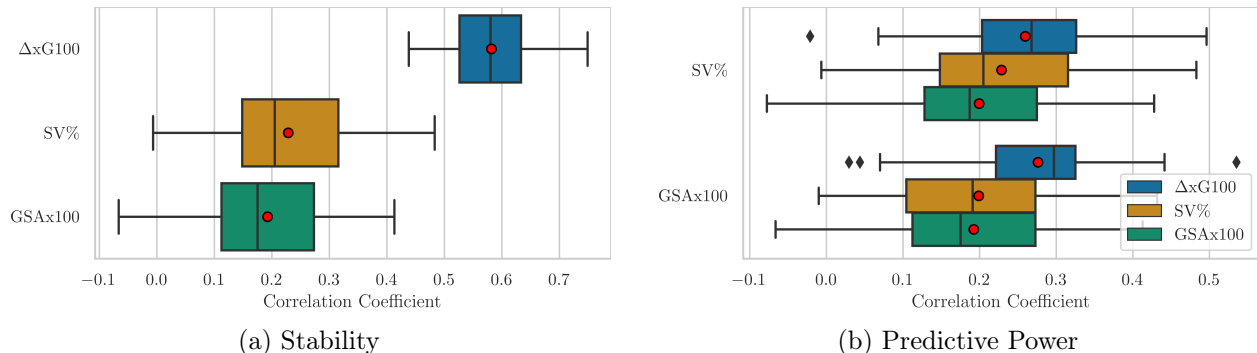


Figure 2.15: The results of the split-half reliability test. The box plots show the distribution of Pearson correlation coefficients across 50 random splits of the data. The red marker shows the mean of the distribution. The proposed metric, $\Delta xG100$, is more stable across the two halves, and better predicts out-of-sample performance.

2.5 Graph-Based Approach

In Section 2.4, effective feature engineering techniques are employed to extract meaningful information from tracking data, resulting in the successful training of a model for estimating expected goals. In this section, an alternative approach is proposed, wherein neural networks are used to train a model that estimates goal likelihood by directly utilizing the instantaneous state of players on the ice at the moment the shot is taken. This approach is motivated by the challenge of manually crafting features to capture complex concepts such as traffic and pressure. Neural networks can learn these concepts directly from data, thereby mitigating the introduction of biases that may arise from manual feature engineering.

This approach is not without its own challenges. Traditional neural networks are not a good fit for this problem as they require a fixed number of inputs and these inputs need to have a natural ordering. The problem at hand has neither of these properties. Penalties in hockey mean that the number of players on the ice is constantly changing. This could be addressed by training a different model for each strength state. This is prohibitive at this moment due to the size of the available sample. A traditional neural network would also require the input to be ordered, which is not the case here. There is no fixed player identity or role that can be used. Thus, the embedding of player states must be invariant to the ordering of the players.

Graphs present a solution to both these issues. Graphs are a mathematical structure that consist of a set of nodes and a set of edges. They are often denoted as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the set of nodes and $\mathcal{E}$ is the set of edges. In the application presented here, graphs can be used to encode the instantaneous game state, where each node represents a player on the ice and the edges represent the relationships between the players. Graph neural networks (GNN), a class of neural networks designed to operate on graph-structured data, can then be used to learn the nonlinear mapping between the instantaneous game state and the goal likelihood of a shot. GNNs have previously been used in the context of sports analytics to predict the yards gained by a rusher in football [55], determine the phases of play in soccer [56], identify tactical patterns in soccer through semi-supervised learning [57], and investigate counterattacks in soccer [58]. They have never been used as the basis for an xG model.

2.5.1 Methodology

The game state at the time of each shot is encoded in a graph. One node in the graph represents the puck while the remaining nodes represent players. Additionally, each node is associated with a node feature vector, which encodes information about the node. The node features are described in Table 2.5. The graphs are fully-connected, meaning each node is connected to every other node. This is consistent with the idea that each player on the ice has full information about the state of every other player. A single edge feature is used, which is the distance between each pair of nodes. At minimum, the graph must have one node for the puck and one node for the shooter. This can be extended to include nodes for the puck and all the players on the ice.

Feature	Description
Location	Location of the entity.
Velocity	Velocity of the entity.
Distance to Offensive Goal	Distance between the entity and the offensive goal.
Distance to Puck	Distance between the entity and the puck.
Angle to Offensive Goal	Angle between the entity and the offensive goal.
Is Shooter	Binary feature indicating whether the entity is the shooter. Set to -1 for the puck.
Is Offense	Binary feature indicating whether the entity is on the offensive team. Set to -1 for the puck.
Is Goaltender	Binary feature indicating whether the entity is the goal-tender. Set to -1 for the puck.

Table 2.5: Node features used to encode the instantaneous game state using graphs.

A network architecture based on Graph Attention Networks [59] is used here, illustrated in

Figure 2.16. The network consists of a series of graph attentional layers followed by a global mean pooling layer. The output of the global mean pooling layer is fed into a fully-connected layer with a single output. In training, a dropout layer is included before the fully-connected layer to prevent overfitting. The number of graph attentional layers, the number of hidden units in each layer, and the dropout rate are considered model hyperparameters. The GNN is implemented using PyTorch Geometric [60].

The loss function is given by (2.5), and is identical to the loss used to train the conventional feature-based model. The model is trained using the Adam optimizer [61]. An early-stopping procedure is used to prevent overfitting. Training is stopped if the validation loss does not decrease for 5 epochs. The previously mentioned hyperparameters, along with the learning rate and batch size, are tuned using Tune [62], and are discussed further in Section 2.5.2.

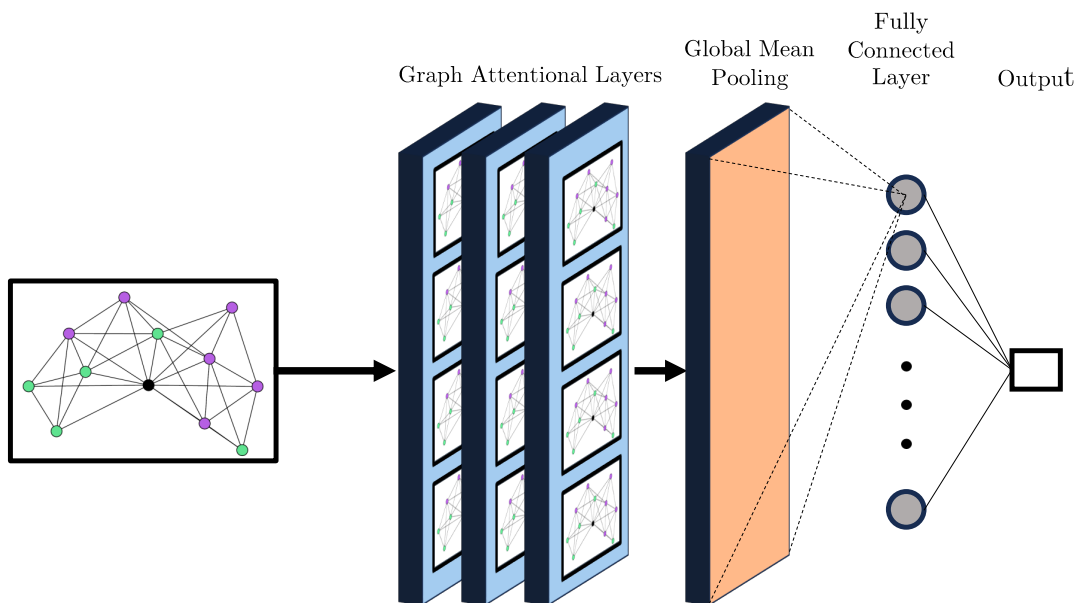


Figure 2.16: Diagram of Graph Attention Network used to learn the nonlinear mapping between the instantaneous game state and the goal likelihood of a shot.

2.5.2 Ablation Study

An ablation study is performed to evaluate the performance of the proposed GNN. The objective of this study is to determine the impact of including additional players in the graph representation of the game state at the moment the shot is taken on the model’s performance. Various permutations of the graph representation are considered. Each graph contains at minimum two nodes, one for the shooter and one for the puck. Additional nodes are added for the skaters on the offensive team, the skaters on the defensive team, and the goaltender

on the defensive team. Assuming the goaltender on the offensive team has little impact on shot quality, the goaltender on the offensive team is not included in any trial. A total of eight trials are performed. A shorthand is used to refer to the trials. For example, OS-DS-DG refers to the trial where nodes are included for the shooter, the puck, the offensive skaters (OS), the defensive skaters (DS) and the defensive goaltender (DG).

This study has more stringent data availability requirements than the feature-based model, as the data must be available for all players on the ice for each shot. Thus, the number of available shots is reduced from 79694 to 72998, 4.2% of which are goals. Adopting the evaluation method from Section 2.4.2, stratified five-fold cross validation is used. Additionally, hyperparameters are tuned only for the OS-DS graph representation to reduce the computational cost of the study. This resulted in the use of four graph attentional layers, each with 32 hidden units. The dropout rate is set to 0.15, the learning rate to 0.001, and the batch size to 32.

A box plot of the loss on the validation fold for each trial is shown in Figure 2.17, and a summary of the results is shown in Table 2.6. The baseline graph representation, with a node for the shooter and a node for the puck, performed better than a naive model, indicating that the GNN is capable of learning the relationship between the states of the shooter and the puck and the goal likelihood. Substantial gains are seen by including nodes for the defensive skaters. The improvements from the addition of the offensive skaters are much less notable. Intuitively, it is unclear how the addition of supplemental information regarding the shooter’s teammates is to benefit this model. The offensive skaters may be responsible for creating traffic, which is shown to have an impact on goal likelihood in Section 2.4.4. However, this particular model is unable to isolate this effect, potentially due to the small sample size. In contrast to that of the offensive skaters, the impact of defensive skaters on shot quality is much more intuitive. As is discussed in Section 2.4.4, pressure exerted on the shooter by the defensive skaters is a critical component of shot quality. It seems likely that the inclusion of nodes for the defensive skaters allows the model to learn this relationship. Additional factors may be contributing to these results, but the opacity of neural networks makes it difficult to isolate them.

These concepts broadly hold when adding a node for the goaltender, which increased performance. Interestingly, the best performing trial uses only three nodes, one for the shooter, the puck, and the goaltender, but the variability between trials is large. These three nodes clearly contain the most significant information regarding the game state when assessing shot quality. Omitting additional skaters may prevent the model from learning spurious patterns and focus on the critical link between puck, shooter, and goaltender. Overall, this ablation study reveals that embedding the game state in a graph representation is a

promising approach for predicting goal likelihood. However, the sample size is too small to draw any definitive conclusions regarding the potential of this approach for outperforming the feature-based model.

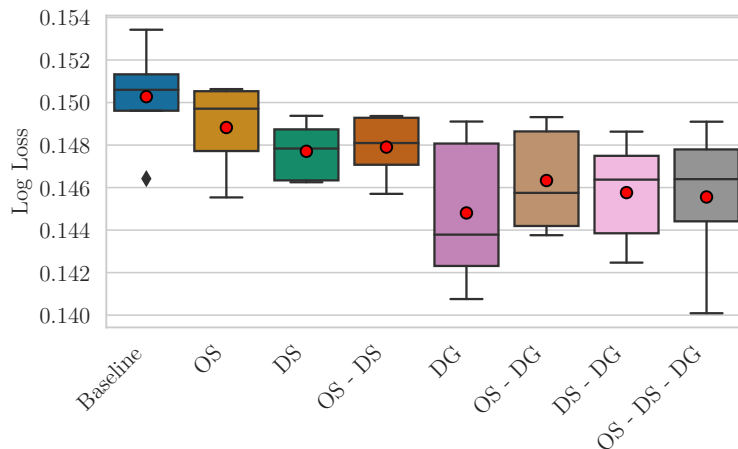


Figure 2.17: Distribution of log loss on validation fold for the GNN ablation study. The trial names are abbreviated as follows: Offensive Skaters (OS), Defensive Skaters (DS), Defensive Goaltender (DG). The trials above the line do not include a node for the goaltender.

Trial	Log Loss	Brier Score	AUC	ECE
Baseline	0.150	0.0376	0.791	4.68×10^{-3}
OS	0.149	0.0375	0.799	4.13×10^{-3}
DS	0.148	0.0374	0.805	3.29×10^{-3}
OS - DS	0.148	0.0374	0.805	3.70×10^{-3}
DG	0.145	0.0365	0.812	3.90×10^{-3}
OS - DG	0.146	0.0370	0.809	3.14×10^{-3}
DS - DG	0.146	0.0369	0.812	3.90×10^{-3}
OS - DS - DG	0.146	0.0368	0.812	3.69×10^{-3}

Table 2.6: Mean value of evaluation metrics on the validation fold for the GNN ablation study. The trial names are abbreviated as follows: Offensive Skaters (OS), Defensive Skaters (DS), Defensive Goaltender (DG). The trials above the line do not include a node for the goaltender.

2.6 Conclusion

This chapter aims to improve expected goals modelling by utilizing tracking data. To do so, a dataset of shots with tracking data is required. This dataset is obtained by adapting the synchronization method from Anzer and Bauer [16] to hockey. Two methods for leveraging tracking data in developing xG models are then presented. A conventional, feature-based

approach is developed first, demonstrating how features can be derived from tracking data to train an xG model. Both the GI-xG and GD-xG models outperform the baseline model, highlighting how tracking data can be beneficial to performance. The features are interpreted using SHAP values, providing insight into the relationship between the proposed features and shot quality. A metric for evaluating goaltenders is also presented, and is shown to be both stable and predictive of out-of-sample performance.

This chapter demonstrates how tracking data can be used to develop a feature-based xG model, but the scope of the proposed model is limited. To prevent overfitting, a rather limited set of features is used. Future work should explore the use of more features, which, with a larger dataset, could improve model performance. Developing independent, data-driven models for pressure and traffic would also be a promising avenue for future work. Lastly, the utility of this model as a performance evaluation tool needs to be further explored, specifically for teams and skaters.

The second main thrust of this chapter is a graph-based xG model. The game state is encoded in a graph, and a GNN is trained to predict a shot’s goal likelihood. The graph-based approach is evaluated using an ablation study, which shows that the proposed GNN is capable of learning the impact of different groups of players on xG, such as the defensive skaters and the goaltender.

As previously discussed, the graph-based approach also suffers from the small sample size. Future work, training on a larger dataset, would be required to determine if this graph-based approach would ever outperform traditional xG models. However, the graph-based approach is a promising one, as it is able to learn concepts such as traffic and pressure from the data, rather than relying on manual feature engineering. In contrast to the feature-based model, the current formulation of the graph-based approach is not provided any temporal information, such as information regarding puck movement. Using a neural network architecture that can capture temporal information, such as Temporal Graph Networks [63], may lead to improved performance.

This chapter demonstrates how tracking data can be used to improve a well-established hockey analytics tool, the xG model. The remainder of this thesis is concerned with developing novel methods that leverage tracking data to explore new or understudied concepts in hockey analytics. This is first done by developing a framework for pass valuation, which is the subject of the next chapter.